Knowledge Network: Faculty & Research

FORAD Reborn: Alumni Duo Saves the Hardest Class You Ever Took

By Samantha M. Novick



Bruce Edlund '96: At Wal-Mart headquarters in Bentonville, Arkansas.



Veselina Dinova '05: At Yahoo! headquarters in Sunnyvale, California.

The first day starts with a warning.

"Welcome to FORAD," Thunderbird Professor Michael Moffett, Ph.D., tells his students. "I want you to know that you don't have to be here. If you can't put the time into this course, or if you are unsure in any way whether you want to be here, I want you to drop today. Your team cannot afford to have you quit on them."

This is Multinational Corporate Finance, a capstone practicum that has tormented Thunderbird students for nearly three decades. "It's a three-credit course that feels more like six credits of work," says Moffett, who has been leading the program for more than 15 years. "FORAD is a job."

At the heart of the curriculum is the FORAD simulation, a game that puts teams of students in charge of their own multinational companies. Over the course of a trimester, each team must make decisions on how to run its hypothetical company.

Players must decide on operations: Production quantity, pricing and the interplay between U.S. headquarters and subsidiaries in Germany and Japan. Then players must determine how to finance their operations. Among key decisions: Whether or not to raise or lower debt, raise or sell equity, transfer funds from the subsidiaries to the holding company and in what currency, and whether or not to hedge.

Teams must submit weekly decisions that represent a 90-day period, one fiscal quarter, in the life of their companies. Moffett then runs the simulation and posts results on the board at the Pub. The company with the highest stock price wins.

But it's still not over. Teams then must complete detailed annual reports and then defend their actions before a panel of alumni and finance faculty. These are the feared Defenses. Overall, the exercise represents about one-third of the FORAD experience.

Different professors — including Brian Heathcotte, Ph.D., James Mills, Ph.D., and Dale Vor der Landwehr,

Ph.D. — have made their mark on the program over the years. New textbooks and other components also have been added, but one thing has remained the same: The FORAD simulation.

Through fall 2011, the basic software remained unchanged from when Thunderbird first licensed it from Professor Lee Remmers of INSEAD in 1982. It ran on MS-DOS with bright Arial fonts on a colored screen with little mouse functionality and no windows.

“It was old when we first used it in ’95,” said FORAD survivor Bruce Edlund ’96, who now works in corporate treasury for Wal-Mart in Bentonville, Arkansas, and returns to Thunderbird twice a year to judge defenses. “When the professor first shows you the model in class and gives you the workbook explaining how to use this program, you can barely believe it.”

In an age when students can trade stocks and manage their bank accounts from their phones, the software might have appeared primitive at best, but it worked … most of the time.

“When you introduce a simulation like this, you give up the ability to really control what happens in your class,” Moffett says. “What happens if it crashes? I’ve been on the phone to Paris at 3 a.m., trying to troubleshoot with (the creator of the simulation) what is going on with the model. I have been really close to just about giving up.”

Updated operating software — specifically Windows Vista and Windows 7 — have made running the simulation even more difficult. But finding an alternative was not an option.

Other financial simulations on the market do not come close to testing the complexities of managing different currencies and financing an international enterprise with as much detail. And no one, including the INSEAD professor who developed the software, was ready to put in the work to upgrade the simulation.

Moffett knew the time had come for someone else to take over the program. Five years ago, after judging another round of defenses, he pulled Edlund aside. “I don’t know what we’re going to do about this model,” he said. “It’s not very reliable. I don’t know how long it’s going to work.”

When Edlund got on the plane to return to home, he started programming. “I guess it’s just in my blood,” he says.

For the next five years in his free time, and for no compensation, Edlund has been building the next generation of the FORAD simulation. From nearly the beginning, he has relied on the experience of alumna Veselina Dinova ’05 to help test and troubleshoot different iterations of the model.

Dinova, a native of Bulgaria who now works in corporate treasury for Yahoo! in Sunnyvale, California, is an equally supportive FORAD advocate. In addition to helping Edlund create the new model, both return to campus twice a year, every year, to help judge defenses. She takes vacation days, travels with her calculator and goes through huge stacks of biennial reports in an effort to give back to the class and the school that changed her life.

“I really appreciated all of the things I learned in FORAD, and as an alumna I want to be able to give back,” Dinova says. “I absolutely love the class, and think it is unique at Thunderbird. Even though I come back here as an expert to judge these defenses, I feel like I continue learning.”

A Thunderbird curriculum without the FORAD simulation is unthinkable to Edlund and Dinova, who both

were part of winning FORAD teams in their respective trimesters. Edlund was a member of the Frog Soup team that eked out a legendarily narrow victory over their French rivals, Frog Legs, and Dinova was part of Fifteen Minutes of Fame.

It is this spirit of competition that FORAD survivors cherish when they recall their experiences.

“The value of the course comes from the intensity and the competitiveness,” Moffett says. “I could stand up there and teach for 10 hours straight, but it wouldn’t come close to the knowledge these students gain from working within their teams, arguing over decisions and coming to their own conclusions together. The model is there to stimulate discovery. As a professor, that’s a dream. It’s what you’re looking for.”

According to Edlund and other graduates, the course itself is a kind of finance status symbol. “Whenever you meet another T-bird alumnus who is in finance, the first question is, ‘Did you take FORAD?’” he says. “And you either did or you aren’t really a finance person. And then the next question always is, ‘How’d you do?’ People go to great lengths to try to win, so that’s a big deal.”

Battle scars

The sheer amount of out-of-class hours required to prepare for complex weekly decisions has the effect of bonding teammates together. “Team dynamics are really important,” Dinova says. “You hope people are going to be reliable and deliver what they promise, and team leaders need to create an environment where everyone is going to contribute.”

The diverse teams also force players to test their cross-cultural skills. “Our team was comprised of five members from five different countries, four different native languages and three continents,” Alberto Cruz ’94 says.

Dominique Quere ’93 still keeps in touch with three of his FORAD teammates. “The key to our success was climbing a nearby mountain,” he says. “Then on the way back, cooking some frozen chicken on a camp fire, trying to eat it while deciding on our strategy for the week to come.”

Moffett also recalls horror stories of teams that couldn’t hold together. “One time I was called by campus security on the weekend because a FORAD team was screaming at each other over a decision in one of the Snell classrooms,” he says.

Other teams have been derailed at the last minute. Rrahul Dalmia ’10 was part of the Cheetahs, which held steady in first place until a disastrous decision in the last period sent them from first to seventh place. “Fatal,” he recalls.

Philip Duncan ’84 remembers a more cutthroat experience. “We had a great run and took the weekly prize for 10 weeks,” he says. “Then we were bumped into second place in the final week after a rival group enlisted a previous FORAD classmate.”

Win or lose, nearly all participants describe the course as a life-changing experience. Robert Birch ’86 says FORAD humbled him and caused him to rethink his career.

“It is the most conceptually difficult class I have ever taken, including law school,” he says. “FORAD is serious business. The acronym strikes fear in the heart, but I would like to take it again.”

Sometimes it is speculation that thrusts a team into first or rips it apart. Moffett warns teams before the game begins that if they speculate wildly — or if they can't defend their decisions at the end — they will lose points. One team in 1995 even made it a running joke, naming their company SpecaSanon, short for *Speculators Anonymous*.

For teams accused of too much speculation, the final defense can turn brutal. "The lights in the auditorium felt like the African sun that night," Jonathan Pearson '00 says. "There must have been 300 people watching me sweat."

The experience was just as traumatic for Julio Benitez '89, who took the course with Professor Mills. "We had heard that every semester one team during the defense would just get pounded," he says. "Well, that semester, it was us. After sitting there for what felt like a lifetime, the panel let us go. All the other teams came up to us to thank us for being the sacrificial lamb for the semester."

Some teams have taken drastic measures to impress the judges. Karen Keese '86 and her team rented tuxedos for their winning defense. David Adam '01 and his winning team enlisted a classmate dressed as a chicken to try and divert the judges' attention away from an uncovered currency hedge that had gone in their favor.

"It didn't work," he says. "I learned an important lesson from that presentation and Moffett's searing glares: Always admit your mistakes and be grateful that sometimes you get lucky. Then, move on!"

One group even tracked down Moffett's wife, ringing the doorbell to get an interview for a video segment that opened their defense. In the video, an interviewer quizzed her on how much her husband loved the group's product, "Arrogance" cologne.

For some teams, Moffett says, the experience is never over. "I have had teams, years later, call me at home on a Saturday morning from another continent after a reunion to say they want to compete one more time — and win this time," Moffett says. "I just tell them to let it go."

Beta testing

"Completing FORAD is almost like a badge of honor," Dinova says. "From my colleagues in treasury, I haven't heard about a course at another school that is remotely similar to this one. The learning experience from a student's perspective is incomparable to any other class because it so closely reflects the real world."

After five years of programming, testing and building, Moffett launched a beta test of the updated FORAD software for the first time in spring 2012. The new interface is clean, sharp and much easier to use. Different windows can be opened for various projections, and teams can see their history of decisions all on the same screen — something impossible with the old model.

The FORAD name and rights also have been taken over by Edlund and Moffett in a partnership with a goal to eventually license the simulation to other schools around the world.

"Vesi and I had such a great experience here at Thunderbird," Edlund says. "And seeing how dedicated Professor Moffett is to the course, how could we not do this?"

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FORAD Survivors Show Off Their Battle Scars

By Samantha Novick



In the hot seat: Students on the Vandelay Industries team make their FORAD defense Dec. 8, 2011. Team members, from left, include Estella He '12, Joseph Hake '12 and Karn Dhandhania '12. Other team members not pictured include Kashisha Dutta '12 and Dave Davidson '12.

Thunderbird School of Global Management students who survive Multinational Corporate Finance (FORAD) love to swap war stories. They laugh about bizarre products such as Sex Panther Cologne — made with real bits of panther. They brag about working 18 hours straight, sleeping on the classroom floor, and then resuming their studies. Perhaps more importantly, they talk about building friendships during the ordeal that have spanned decades. Following is a sampling of FORAD memories shared by T-bird alumni. Responses came from Twitter, Facebook, LinkedIn and e-mail.

Philip Duncan ’84: Working until 2 a.m.

We had a great run in ’84 with two partners (Dave Seed ’84 from Canada was a star), and took the weekly prize for 10 weeks. We were bumped to second in the final week after a rival group enlisted a previous FORAD classmate, but we had a great time, including daily time working until 2 a.m. on the computer simulations. Professor was also great.

Philip Duncan ’84: A very low curve

Our professor for ‘84 FORAD was Paul Dickie, and he made the course very enjoyable in his own style, including sophisticated case study exams with a very low curve. If anybody knows where he is now, I would like to contact him as he is not on T-bird search engine.

Judy Canfield Power ’84: Missed opportunity

I remember that we didn’t buy back our stock when we should have, and it cost us first place.

Silvana Roncal ’85: The program ran on DOS!

It would be very interesting to compare the FORAD of 1985 to the current one. Back in my time, my team won and I later became the TA for the game. It was a sophisticated treasury operation within a multinational dealing with cash management, translation and liquidity issues in more than two currencies. The program ran on DOS!

Kristi Buckles Otto ’85: Beware of saboteurs

I agree with Silvana. It would be great to compare the 1985 version versus today. Our group spent what

seemed like a a great deal of time keeping another group from sabotaging our programs! I really enjoyed the INTERAD program.

Karen Keese ’86: Still friends to this day

I took FORAD in fall 1985 with Dr. Dickie. Anthony Hull ’86, Robin Carpenter ’86 and I were a three-person team, and are still friends to this day. We won that semester, got a bit punchy, and rented tuxes for the defense. I think I still have the photo. We were all FORAD TAs the following semester. Ant had the brilliant finance mind, Robin the diligence to read and review everything, and I was the risk mitigator. I was extremely fortunate with my partners.

Robert Birch ’86: Five sharpened pencils in my shirt pocket

FORAD was a profoundly humbling experience that convinced me I would not make it in investment banking and did not create feelings of self-assurance during my on-campus interview with Morgan Stanley. In early ’86, our handsome professor, a University of Southern California Ph.D. and former Bank of Montreal wizard, was in his last term before going to the Asian Development Bank. Intuitively, he could solve any finance problem and write the equation on the blackboard but could not explain how he came up with the solution. Few of us could follow his advanced financial logic and sat in class wondering how he got from the problem set to the elegant solution with so few calculations. Explanations ruined the artistry and Zen-like wizardry. He tried explaining a couple of times and the conceptual baby steps necessary for the whole class to understand drove him into temporary madness, as there was more than one way to solve the same problem, take your choice of methods. We must find our own solution, how we arrived didn’t matter much. We learned to shut up and watch carefully. The weekly projects took me 40 grueling hours to solve on Lotus 1-2-3 and write up on WordPerfect 4.2. Walking to the final examination I had five sharpened pencils in my shirt pocket and plugged my ears to seal my brain and remember foreign currency translation equations. Final exam included a shrimp processing facility project in the Philippines owned by a Hong Kong parent company with an American holding company — determine pre and after tax IRRs at the local, parent, and holding company level. I completed about two-thirds of the final exam case and survived the course with a B. I waived Mandarin Chinese, and FORAD is the most conceptually difficult class I have ever taken including law school — but would like to take it again. FORAD is serious business; the acronym strikes fear in the heart.

Alex Blochtein ’88: A New York law firm

FORAD was by far the best class at Thunderbird. I have outstanding memories of this class. James Mills was a great professor and a real character. Our team excelled. Many friends used to call our team a “New York law firm” as our last names are: Bletchman, Blochtein and Borjeson. I still use some of the techniques to minimize exposure. It was a true learning experience.

Julio Benitez ’89: The sacrificial lamb

Forad was wonderful. I was fortunate enough to not just take the course but was then the T.A. the following semester. Dr. Mills was wonderful. My funny story (funny now) relates to the defense. We had heard that every semester one team during the defense would just get pounded. Well, that semester it was us. After sitting there for what felt like a lifetime, the panel let us go. All the other teams came up to us to thank us for being the sacrificial lamb for the semester. Thanks Tracey and Pradeep.

Ron Lagraff ’92: ‘Are you on drugs?’

I took FORAD with Dr. Dale (not "Dave") in 1992, and it was definitely the best finance class I ever took. A really funny exchange happened during our defense in front of a group of professors and a private sector banker. Professor Frank Tuzzilino was on the panel, and when it became his turn, he asked a question that had a cryptic introduction about "northern and southern hemispheres" of the balance sheet. I remember Dr. Mills taking his glasses off, staring him down and asking, "Frank, are you on drugs?" It was a very funny moment in what was (for us anyway), a tense exercise. Interesting that, after so many long hours and long nights, it's this little exchange during our defense that I remember most all these years later.

Paul Tsakiris '92: Truly exceptional courses

I thought it was a great class. My only regret is that I took it over the summer and was not able to focus on it as much as I would have liked. Definitely the best finance class I have ever had! FORAD and INTERAD are truly exceptional courses.

Jonathan Pearson '92: The heat of the African sun

We were quite confident in the second week's decision-making and came out way WAY ahead. Calling it speculating might be a stretch, but our panel certainly didn't think so. The lights in the auditorium felt like the African sun that night. There must have been 300 people there watching me sweat.

Dominique A. Quere '93: Chicken on a campfire

FORAD brings back some good memories back in the spring of '93 … still in touch with three of my four teammates from those days. The key to our success was climbing a nearby mountain, then on the way back, cooking some frozen chicken on a campfire, trying to eat it and deciding our strategy for the week to come.

Mary-Jeanne Cabanel '93: A FORAD superstar

I remember that a FORAD superstar worked with me all night to save my entrepreneurship project that almost went bankrupt before it started.

Alberto Cruz '94: Five members from five countries

I had an outstanding FORAD experience. Our team was comprised of five members from five different countries, four different native languages and three continents. We screwed it up big time trying to speculate big time, just for once, and had a terrible time justifying some of our consistently inconsistent decisions.

Scott Hessell '96: Failing with Dr. Moffett

My experience was nowhere near as good as others. Our team, quite frankly, did horribly. But I remember Dr. Moffett telling us in class — and I believe personally too— that he would rather we fail with him than fail on the job. Now, among many things that I'm doing, I teach part-time at the University of Arizona, and I tell my students the same thing all the time and have structured many activities with that in mind. And, based in no small part on my experience in FORAD, I'm also starting a company with former Deloitte application developers that is creating a series of competitive business simulations for the retail marketplace to be used at the university and professional levels. So, while my memories are not as fond as for those who won, I think I may have learned just as much, if not more so.

Andreas Kraemer '96: Speculating like there was no tomorrow

I remember that we speculated like there was no tomorrow during the last decision, and that we got our behinds kicked thoroughly during the defense because of it.

Janet Marie Joyce ’97: Power pink suit

I remember my power pink suit. Hello to Professor Moffett!

Julianne Ferencz Babione ’97: The wall at the Pub

I remember checking the weekly results posted on the wall at the Pub.

David Adams ’01: Sometimes you get lucky

Our group won in the spring of ’01 based on an uncovered currency hedge that went in our favor. We had a friend march in during our introduction dressed up in a chicken costume as our mascot in an effort to divert the attention from our dumb luck. It didn’t work! I learned an important lesson from that presentation and Moffet’s searing glares: Always admit your mistakes and be grateful that sometimes you get lucky. Then, move on!

Douglas Kennedy ’03: Pushed to the limit and then some

My research during FORAD into international tax and advanced cost accounting was very insightful. Have you ever wondered how it is the big multinationals, such as GE, pay no federal tax? That is four courses by itself. I loved FORAD because it pushed me to the limit and then some. When Big Daddy Jim Mills and Moffett taught Financial Engineering and FORAD , a student was getting a tremendous education. It bought you the first 90 days at one of the big firms. My two regrets are that I didn’t get to take INTERAD, and I didn’t complete a dual language thesis. FORAD — simply the best course I’ve ever taken.

Jose Dominguez ’04: Physical Restraint Devices

I remember people laughing about our company name: Physical Restraint Devices … and people laughing even more about our lame defense!

Nikkole Bonjorni ’09: 60 percent of the time it works every time

I think there are many skills you learn in FORAD that can be applied post-graduation and not all of them are finance related. I think the main thing is learning to work in a team under stressful conditions. While you are working you know that any decision made within the team may have to be defended in front of your peers. I think, in most cases, it makes people more cautious and willing to listen to other people. That is a priceless skill to learn. I lucked out because my team got along well with one another. Of course, we were a team of four women and one man … which makes for a less stressful scenario if you ask me. ;) The results were always posted on Thursday. Each week we had a decision due. We would meet Thursday night to discuss our results (where we went wrong, what strategy we wanted to take, etc.) over dinner. No laptops. Just discuss. I really think it helped us to analyze what happened. I think it helped keep the mood light so we didn’t get aggravated with one another later on. It happened, of course, but we didn’t resent anyone or despise anyone after the class was over. My group did not win best biennial but we did get an honorable mention for most entertaining. I think that says something about how our group worked together. :) Our product was Sex Panther Cologne … made with real bits of panther. Our slogan: “60 percent of the time it works every time.” It was a good memory. I am not a finance major and I don’t work in finance. In fact, finance is probably my

worst subject, but I took this class because I wanted to strengthen my knowledge in that area. And this class did just that.

Rrahul Dalmia ’10: Dropping from No. 1 to No. 7

FORAD was an amazing overall business tool. I remember a steep learning curve, sleepless nights at the Commons and TSG. I remember our fatal mistake that dropped our rank from No. 1 to No. 7. At the end of it, you wish you could play “just one more time.”

Nathan Skon ’10: Breaking into the field

I loved the course, and it directly got me into the field into multinational finance, and I now manage the foreign exchange program at SunPower.

Manish Pandey ’10: Me and three Chinese

I was in the FORAD class of spring 2010 and I had three Chinese ladies in my group. It was probably the best college experience I had in spite of ending at ninth place!

Amitesh Kumar ’11: Sleeping on the classroom floor

I took FORAD in Spring 2011 and it was one of my favorite courses while at T-bird. One of the reasons is the method adopted by Professor Moffett. It definitively improves one’s financial management skills in a global context, but along with that, it helps one develop leadership and team skills. Some Memories: 1) We had been working on biennials (Antoine, Ashustosh, Subu and myself) for almost 18 hours at a stretch, just before submission day, and I remember Antoine being so tired that he just slept on floor of Snell 4, where we used to have team meetings, but then once he woke up he was back to work. 2) We never had arguments within our team, and this helped us listen to others, which many employees of companies don’t do these days. 3) The results of each period used to be displayed on Monday evenings in the Pub, and we used to sit as a team sipping beer waiting for result because it’s addictive. It got all of us involved, and the best thing is we considered it as running a company rather than just a game. 4) We were ranked second in defense and third best biennial. So, in short, it was mixture of learning, winning and fun